Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal force” and “tension” to explain your model or result. Sentence starters are optional.

normal force: a support force perpendicular to a contact surface

tension: the pulling force through a stretched rope or cable

weight: the gravitational force on an object

01 Bring back an earlier idea

A fixed-mass cart has acceleration 3 m/s² under 6 N net force.
Find acceleration when net force becomes 10 N.
Explain.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = 9.8 N/kg unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight 12 N down, the cable tension is 12 N up. Only forces acting on the lamp belong in its diagram.

Your turn: Draw the lamp as a labeled point with its two force arrows.
Explain why its force on the cable is not another arrow on this diagram.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 A book at rest

Draw and label the forces on a book resting on a level table.
Name both sources.
Use arrows whose comparative lengths match rest.

Sentence starter: The … in my model represents … . Its label shows … .



Add and label force arrows.

The dot represents the object. Object: book

Marzano check - circle: ① ② ③

04 After the hand lets go

A box slides to the right on a rough floor after release.
Draw its forces. Which way does friction act?

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A falling ball

A ball falls down and is speeding up. Air pushes up. **Draw** weight and air resistance. Which arrow should be longer?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3

06 Choose a method

A cart goes from 18 m to 6 m in 4 s at steady velocity. **Find** velocity and sketch its position-time graph.

Word help: velocity = rate of position change, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A ball moves upward after leaving a hand. Ignore air resistance.
Draw its free-body diagram and justify every arrow.

Word help: justify = support your answer with a reason or evidence

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Draw separate diagrams for a hanging lamp at rest and the same lamp moving upward at steady speed. What can a force diagram alone fail to tell you?

Word help: at rest = not moving in the chosen reference frame

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3